

ThesisForge

Multi-agent thesis quality control

A Framework for Sustainable Urban Mobility: Integrating Policy, Behavior, and Technology

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Doctor of Philosophy

May 2025



Literature Review

3.2 Empirical Evidence

Methodology Overview



References

Zhang, Y. (2022)	Transportation Research Part D	✓
Litman, T. (2021)	Victoria Transport Policy Institute	✓
Banister, D. (2008)	Transport Policy	✓
Cervero, R. (2013)	Journal of Urban Planning	⚠
Bibri, S. E. (2019)	Sustainability	✓

Similarity Check

Overall Similarity **12%**

Within acceptable limits



Thesis Health Report

Overall Quality ⓘ

86

Good

Originality

82

Good

Methodology

88

Good

Rigor & Validity

84

Good

Clarity & Coherence

87

Good

Citation Check

Total References	128
Verified	112 87%
Partial Match	9 7%
Not Found	7 6%

[View Citation Report](#) →

Priority Fixes **7**

- High**
Clarify research gap and motivation
- High**
Strengthen methodology justification
- Medium**
Address limitations in model assumptions
- Medium**
Add robustness check for key results
- Low**
Improve transitions in Chapter 4

[View All Issues](#) →

Defense Questions **5**

- How does your framework compare to existing models?
- What are the limitations of your data sources?
- How do you justify the choice of independent variables?
- What are the policy implications of your findings?
- How would you extend this research in the future?

[View All Questions](#) →

Report Summary

Good overall quality with solid methodology and clear structure. Address the high-priority issues to strengthen rigor and impact.

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Agent Orchestrator

Routes tasks and synthesizes insights ✓



Literature Agent

Evaluates sources and research context ✓



Methodology Agent

Checks design, rigor, and reproducibility ✓



Analysis Agent

Validates results and statistical soundness ✓



Writing Agent

Reviews clarity, flow, and academic tone ✓



QA Synthesizer

Aggregates findings and recommends actions ✓



Multi-agent quality control

Comprehensive thesis review across all critical dimensions



Evidence-based insights

Actionable feedback with clear prioritization



Academic integrity first

Citations, originality, and rigor you can trust



Defend with confidence

Better answers. Stronger research. Impact that lasts.

Thesis issues are found too late.

Drafts, references, results, supervisor feedback, and defense prep usually live in separate tools. Students often assemble the thesis before discovering whether the work is coherent and defensible.

- ! Research gaps are broad, outdated, or weakly supported.
- ! Claims drift away from the uploaded references.
- ! Methods and results do not always match the stated objectives.
- ! Defense preparation starts after the risky parts are already written.

A quality-control workspace for early-stage researchers.

ThesisForge reviews the work users already have. It turns thesis text, references, result files, and project metadata into a structured report with scores, priorities, citation notes, and defense questions.



Thesis Health Report

Overall quality score with visible evidence and next actions.

Priority Fixes

Ranked work items for the highest-risk sections.

Citation Checks

Claim support, partial matches, and missing references.

Defense Questions

Likely panel questions with answer guidance.

Action Tasks

Concrete follow-up tasks users can track.

Seven agents review the thesis from different angles.

The workflow is visible: users see progress, handoffs, collaboration messages, and how the final report was assembled.

1

Literature

Coverage, themes,
missing synthesis.

2

Research Gap

Specificity, currency,
defensibility.

3

Citation

Claim-to-reference
alignment.

4

Methodology

Problem, objectives,
method fit.

5

Results

Interpretation and
overclaiming checks.

6

Defense

Panel risks and answer
points.

7

Report

Scores, fixes, tasks,
summary.

Judges can test the full workflow quickly.

The app includes a safe synthetic thesis project so reviewers can load data, run the agent workflow, inspect collaboration logs, and open the final report without using private academic material.

1. Load demo project

Creates project metadata, thesis text, references, and result files.

2. Start thesis review

Queues an asynchronous analysis run through Redis/RQ.

3. Watch agent progress

Timeline and collaboration log make the review traceable.

4. Open final report

Scores, fixes, citations, and defense prep are ready to review.

Built as a real full-stack product.

The MVP has authenticated project workflows, parsing, background jobs, agent orchestration, report persistence, deployment config, and a safe analytics path that avoids storing private thesis content.

Next.js

React

TypeScript

Tailwind CSS

FastAPI

Pydantic

SQLAlchemy

Alembic

Postgres

Supabase Auth

Redis

RQ

OpenAI

Band

Docker

Vercel

Render

5

file types supported for MVP uploads

7

review agents in the workflow

1

synthetic demo path for judges

0

private thesis text in analytics

Better thesis review before submission.

ThesisForge gives students a clear path from scattered materials to focused revision work, while keeping the product boundary on review, evidence, and preparation.

Academic integrity first: review the thesis, surface risks, and help the student defend stronger work.

- ✓ Prioritized fixes reduce last-minute uncertainty.
- ✓ Citation notes help strengthen evidence and rigor.
- ✓ Defense questions turn vague anxiety into preparation.
- ✓ Visible agent handoffs make AI-assisted review traceable.